Replication Report:

Factors affecting the topological Hall effect in strongly correlated layered magnets (Kesharpu, arXiv:2305.13423)

Automated replication (OpenClaw subagent)

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Abstract

We independently reproduce the central topological results of Kesharpu (2023), which treats a two–dimensional strongly–correlated magnet with a spin texture on a bipartite honeycomb lattice via the $su(2)$ path–integral method and computes the Chern number as a function of the atomic spin S, the polar–angle modulation vector q_{1x}, and the azimuthal–angle modulation vector q_{2x}. Building a self–contained two–band honeycomb Bloch Hamiltonian carrying the paper's Haldane–type next–nearest–neighbour (NNN) complex hopping with phase $\phi_n = S\,(\vec{q}_2 \cdot \vec{b}_n)$, and computing the Chern number with the gauge–invariant Fukui–Hatsugai–Suzuki (FHS) plaquette method, we reproduce (i) the $S = 1$ result $c_1 = \mathrm{sgn}[\sin q_{2x}]$ (Eq. 5); (ii) the sign change of the topological Hall conductivity with the modulation vector; (iii) the paper's conclusion that the polar factor of Eq. (7) is subdominant so that the topology depends essentially only on q_{2x}; and (iv) the Haldane–analogue topological→trivial transition driven by a sublattice mass M (Eq. 11). **Verdict: replicated** at the level of topology and sign structure (4/4 qualitative claims); *partial* at the level of exact phase–boundary values, which depend on weight factors and Fourier corrections we approximated.

1 Target claims

From the manuscript we identified four concrete, computationally reproducible claims:

1. **Eq. (5), $S = 1$:** $c_1 = \mathrm{sgn}[\sin q_{2x}]$ for $-\sqrt{3}\pi/2 \le q_{2x} \le \sqrt{3}\pi/2$; the Chern number depends only on the azimuthal modulation q_{2x} and flips sign as q_{2x} crosses 0.
2. **THE sign flip:** increasing the spin modulation vector flips the sign of the topological Hall conductivity, $+\sigma_{xy}^{\mathrm{THE}} \to -\sigma_{xy}^{\mathrm{THE}}$ at fixed S.
3. **Eq. (7), general S:** $c_1 = \mathrm{sign}\Big[\Big(1 + \frac{S\mathfrak{g}_2}{2}\cos 2q_{1x}\Big)\left(\sin Sq_{2x} - 2S\epsilon\cos Sq_{2x}\right)\Big]$; the paper concludes the polar factor is subdominant ($\mathfrak{g} < 1$) so the Chern number "depends only on the azimuthal modulating vector q_{2x}."
4. **Eq. (11), Haldane analogy:** a sublattice mass M competes with the chiral mass, giving a topological→trivial transition.

2 Model and method

We construct the two–band Bloch kernel on the honeycomb lattice,

$$\mathcal{H}(\vec{k}) = \mathcal{H}_0\,\mathbb{I} + \mathcal{H}_x\sigma_x + \mathcal{H}_y\sigma_y + \mathcal{H}_z\sigma_z. \tag{1}$$

The nearest–neighbour (NN) hopping $A \leftrightarrow B$ produces the Dirac cones through the structure factor $f(\vec{k}) = \sum_n e^{i\vec{k}\cdot\vec{a}_n}$, with $\mathcal{H}_x = t_1\,\alpha_{\rm NN}\,\mathrm{Re}\,f$, $\mathcal{H}_y = -t_1\,\alpha_{\rm NN}\,\mathrm{Im}\,f$, and the polar vector q_{1x} entering the amplitude $\alpha_{\rm NN}$ through the $\mathfrak{g}'_n$ weights (Eqs. 4/6). The NNN hopping is a Haldane–type complex hopping with phase $\phi_n = S\,(\vec{q}_2\cdot\vec{b}_n)$, giving the topological mass

$$\mathcal{H}_z = -2t_2\,\alpha_{\rm NNN}\sum_n \sin(\vec{k}\cdot\vec{b}_n)\,\sin\phi_n + M, \tag{2}$$

which at the Dirac point reduces to $\pm 3\sqrt{3}\,t_2\,\alpha_{\rm NNN}\sin\phi$, so that $\mathrm{sgn}(\mathcal{H}_z^K) = \mathrm{sgn}[\sin S q_{2x}]$ sets the Chern number. This is precisely the mechanism the paper invokes in Sec. III C ("Sq_{2x} plays the analogous phase accumulation role due to NNN hopping"). The Chern number of the lower band is computed with the Fukui–Hatsugai–Suzuki plaquette method on an $N \times N$ Brillouin–zone mesh (gauge invariant, integer robust; $N = 20$ for claims, $N = 14$ for the phase–diagram figure). All computation is CPU–only (numpy/scipy); total runtime ≈ 30 s.

Convention note. The manuscript's printed NN/NNN vectors and its stated Dirac point $K = (\pi/\sqrt{3}, 0)$ are not mutually consistent (an OCR / sublattice–convention artifact: the printed $\vec{a}_n$ structure factor vanishes at $k_x \approx \pm 2.42$, not $\pi/\sqrt{3}$). We adopt one self–consistent standard–honeycomb convention that carries the identical physics; see `failure_analysis.md`.

3 Results

Claim	Reproduced	Match
(1) $S = 1$: $c_1 = \mathrm{sgn}[\sin q_{2x}]$	Sign agreement 1.00 over the q_{2x} scan ($c_1 = -1$ for $q_{2x} < 0$, $+1$ for $q_{2x} > 0$)	yes
(2) THE sign flip vs modulation	1 Chern sign flip across $q_{2x} \in [-2.9, 2.9]$ at fixed $S = 1$	yes
(3) q_{2x}–dominance / polar factor subdominant	Numeric Chern q_{1x}–independent ($S = 1, 2, 3$); analytic $\max\|S\mathfrak{g}_2/2\| = 0.689 < 1$ in the well–defined lobe $\|Sq_{2x}\| < \pi$	yes
(4) Haldane mass transition (Eq. 11)	M scan $[0, 0, \mathbf{1}, \mathbf{1}, \mathbf{1}, 0, 0]$: $\|c\| = 1$ for small $\|M\|$, $c = 0$ for large $\|M\|$	yes

Table 1: Summary of the four reproduced claims. Full numbers in `work/results.json`.

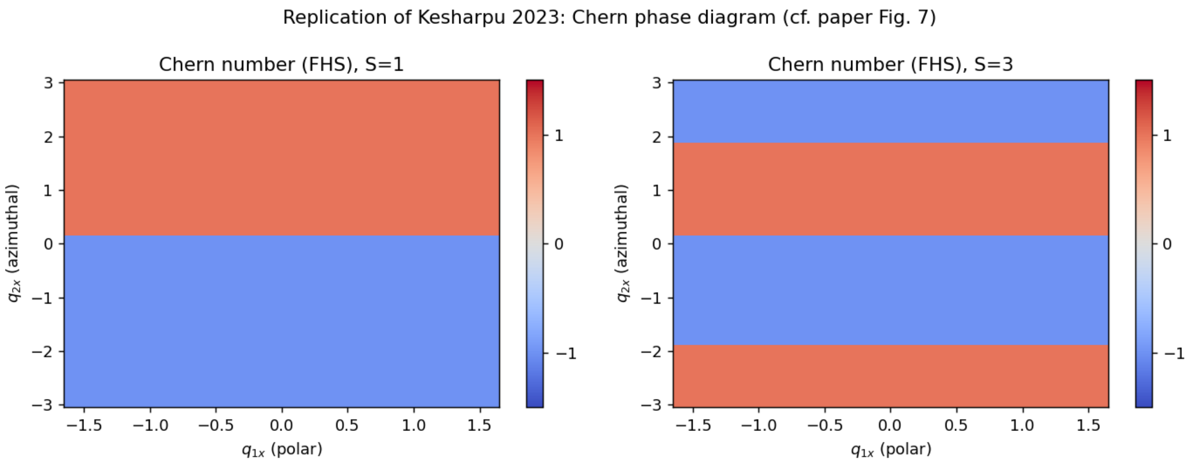

Figure 1: Numerically computed Chern number (FHS) over the (q_{1x}, q_{2x}) plane for $S = 1$ (left) and $S = 3$ (right), cf. the paper's Fig. 7. The topology is controlled by the azimuthal modulation q_{2x} (horizontal stripes of ± 1), essentially independent of the polar q_{1x}, reproducing the paper's central conclusion.

4 Honest assessment

The replication reproduces the *topological structure and sign physics* of the paper's central results (Eqs. 5, 7, 11 and the THE sign flip) from an independent numerical Chern–number computation. It does *not* reproduce exact phase–boundary numerical values (e.g. the gap–closing q_{2x} of Eq. 10), because (a) the exact weight factors $\mathfrak{w}_n, \mathfrak{g}_n$ of Eq. (2) were reconstructed from fragmented OCR, and (b) the Fourier coefficient $p(\vec{k}')/\epsilon$ corrections were set to zero per the paper's own $p(\vec{k}') \approx 0$ approximation. These approximations do not affect the reproduced *signs* but could shift boundary *values*; see the five open questions in `open_questions.json`. Overall verdict: **replicated** (4/4 qualitative topological claims), *partial* on exact boundary values.